

Boolean & LSI Information-Retrieval Toolkit - Report

1. Introduction

This report details a toolkit designed for information retrieval, leveraging TF-IDF and optional LSI indexing over Amazon review data. It provides functionality for Boolean filtering combined with cosine-similarity ranking to return relevant documents based on user queries. The toolkit is developed with a focus on simplicity and efficiency, utilizing core Python libraries for processing and retrieval.

2. Project Context

This project aims to provide a lightweight and efficient information retrieval solution tailored for text-based reviews. It allows users to perform complex searches using Boolean logic and rank results by relevance, making it suitable for analyzing large datasets of reviews. The toolkit avoids heavy IR frameworks, relying instead on NumPy and pure Python 3, ensuring accessibility and performance.

The project can be found in the following Github repository:

<https://github.com/jeremyelalouf/nlp>

3. Toolkit Components

The toolkit includes two primary scripts:

- ``convert_reviews.py``: Converts the original JSON-lines dump of reviews into a directory tree of plain-text files.

- ``main.py``: Indexes the files (or a JSONL file) on-the-fly and processes Boolean queries.

4. Directory/File Layout

After converting the dataset, a structured file layout is recommended for optimal performance:

```
Unset
reviews_folder/
├── ASIN_NUMBER/
│   ├── 0000000.txt
│   ├── 0000001.txt
│   └── ...
```

Each unique ASIN (Amazon Standard Identification Number) becomes a sub-directory, and each review file contains the "summary + reviewText" of an individual review.

5. Dataset Conversion

To convert the dataset, execute the following command:

```
`$ python convert_reviews.py --input reviews.jsonl --outdir reviews_folder`
```

This command processes the input JSONL file and outputs the converted data into the specified directory.

6. Searching with the Toolkit

Indexing and querying are performed in a single call using the ``main.py`` script:

```
`$ python main.py reviews_folder --query "(knit OR stitches) AND NOT (index)" --top_k 10`
```

Key Arguments

Argument	Description
`reviews_folder`	Directory (or JSONL file) containing the reviews
`--query`	Boolean expression using AND/OR/NOT, with parentheses for complex queries
`--top_k N`	Returns the top N best-scoring documents (default is 5)
`--topics K`	Applies Latent Semantic Indexing (LSI) with K latent topics if >0
`--max_vocab N`	Keeps the N most frequent tokens (trades off between speed and RAM usage)

7. Experimental Results (Sample)

Example results based on a dataset of 5 sample reviews for ASIN 0449819906.

Query

```
`(knit OR stitches) AND NOT (index)`
```

Rank	Score	Snippet
1	0.4227	Contains some interesting stitches.
2	0.3765	...reversible stitch patterns offered and shown reverse and obverse...

Both documents satisfy the Boolean filter, and the cosine scores reflect their relevance within the filtered set.

8. Performance and Dependencies

The toolkit is built using pure Python 3 and NumPy, avoiding any heavy IR frameworks. On approximately 100,000 reviews, index building takes about 10 seconds on an M1 MacBook Air. This emphasizes the toolkit's efficiency and suitability for resource-constrained environments.